

Assignment 1

28/09/2026

100 Points Possible

Attempt 1



In Progress

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Unlimited Attempts Allowed

12/09/2026 to 28/09/2026

▼ Details

Every year since 1946, representatives of the UN member states gather at the annual sessions of the United Nations General Assembly. The centrepiece of each session is the General Debate. This is a forum where leaders and other senior officials deliver statements that present their government's perspective on the major issues in world politics. The 2026 UN General Assembly and General Debate (81st session) is happening during the Fundamentals of Data Science course, exactly when you will be working on this assignment (more info [here \(https://www.un.org/en/high-level-week-2026\)](https://www.un.org/en/high-level-week-2026).)!

Your assignment is to analyse a dataset composed of the corpus of texts of UN General Debate statements from 1946 (Session 1) to 2025 (Session 80) and relate it with other datasets (e.g., the Happiness Report or the International Trade Dataset used in Lab 2). The United Nations General Debate Corpus is available [online \(https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/0TJX8Y\)](https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/0TJX8Y) and you can gain extra insight about its contents by reading [this paper \(https://doi.org/10.1177/2053168017712821\)](https://doi.org/10.1177/2053168017712821) (pdf [here \(https://canvas.uva.nl/courses/59831/files/15868331?wrap=1\)](https://canvas.uva.nl/courses/59831/files/15868331?wrap=1)) or [this one \(https://doi.org/10.1177/00223433241275335\)](https://doi.org/10.1177/00223433241275335).

The goal of this assignment is to perform an exploratory data analysis (i.e., **data collection, data pre-processing, combination of datasets, descriptive statistics, and visualisation**) and predictive analysis (i.e., **model selection, training and validation**). You are expected to apply statistical learning methods and report obtained coefficients and/or validation metrics. You are expected to first **gather information about the dataset** (i.e., read the introductory paper and dataset documentation; inspect some of the entries in the data set; inspect the Happiness Report Dataset, International Trade Dataset or others) and **formulate and answer questions** that you would like to explore.

You should pose **one exploratory** and **one predictive** question. Ideally, the **Exploratory Data Analysis (EDA)** should allow to refine your predictive question. You will be required to **clean** and **pre-process the textual data** in a suitable way. Note that these multiple tasks can inspire an efficient division of labour for each of the group elements to focus on.

The theme for the general debate this year (81th) session of the General Assembly is "*Restoring trust, managing transformation*". Your questions must be related to the theme, **and one (1) Sustainable Development Goal (SDG) (https://sdgs.un.org/goals) of your choice**. While not all

SDG are directly related to this year's theme, many will overlap; sometimes in non-obvious ways. You must make explicit in your report which SDG you picked and the connection with the questions you posed. We strongly suggest that you discuss your question(s) with your TAs before starting your analysis.

To inspire you, here are some suggested question from 2024's theme ("*Leaving no one behind: acting together for the advancement of peace, sustainable development and human dignity for present and future generations*"):

- Is there any apparent correlation between the frequency of poverty-related mentions in political speeches and actual poverty rates over time? (Exploratory question, SDG 1)
- Are there any visible trends in how different continents refer to ocean-related issues in political speeches? (Exploratory question, SDG 14)
- Can we predict the amount of ocean-related products that a country exports based on the text of speeches? (Predictive question, SDG 14)
- Can we predict the number of forest conservation agreements in a continent based on the text of speeches given by countries of such continent? (Predictive question, SDG 15)

The questions above are merely indicative; feel free to come up with the questions and directions that you find most engaging. Please keep in mind that we will reward creativity in the questions proposed (i.e., we will value if you are the only group suggesting some very original question).

The main purpose of this assignment is to evaluate whether you are able to master the first 4 weeks of the course, lectures and practicals. In particular, you will be asked to 1) clearly define which question(s) to explore, 2) pre-process the data, 3) combine multiple datasets, 4) apply the functionality of NumPy/Pandas/Matplotlib/Scikit-learn, 4) generate suitable data visualisations and 5) apply statistical learning techniques.

Datasets

The base dataset to use is the raw text data corresponding to the corpus of texts of UN General Debate statements from 1946 (Session 1) to 2025 (Session 80). More info [here](https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/0TJX8Y) (<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/0TJX8Y>) and [here](https://doi.org/10.1177/00223433241275335) (<https://doi.org/10.1177/00223433241275335>). Besides using the UN General Debates Corpus, you must combine multiple datasets. Note that we expect that, in your questions, you combine data of the UN General Debates Corpus with at least two other datasets. You can use the Happiness Report Dataset or the International Trade Dataset introduced in the Labs, but you are also allowed to combine it with other datasets, *provided they are well-sourced*. [Our World in Data](https://ourworldindata.org/) (<https://ourworldindata.org/>) is a potential source, or at least can help you to narrow-down extra datasets to use.

In Lab 3 you'll also practice how to work with the UN General Debate corpus (see Lab3 Aux file).

Report and Submission

Please submit a .zip file containing a .pdf report and the code (e.g., .ipynb notebook) you used in your analysis. Each group is expected to work independently in their code/notebook. Grading will be mostly based on the report and the code will only be checked in case any doubts arise on how the analysis was performed. *Make sure that your notebook is reproducible and can be run from scratch* easily (this includes public links to the extra raw data used, if applicable). The report should contain all the information for a reader to understand the analysis and results. Plagiarism detection will be used (the '[Regulations governing fraud and plagiarism for UvA students \(https://student.uva.nl/en/information/plagiarism-and-fraud\)](https://student.uva.nl/en/information/plagiarism-and-fraud)' applies to this course. This will be monitored carefully. Upon suspicion of fraud or plagiarism, the Examinations Board of the programme will be informed).

Your report should be ~7 pages. The maximum is 9 pages, not including references (any page beyond the page limit will not be read and will lead to a penalty in the final grade). The report styles are loosely based on the style of computer science scientific journal papers. [The basic guideline for the lay out of your report \(https://canvas.uva.nl/courses/59831/files/15868341?wrap=1\)](https://canvas.uva.nl/courses/59831/files/15868341?wrap=1) is as follows:

Title - Author List (Mandatory, alphabetic order) - Abstract - Introduction - Methodology - Results and Discussion - Conclusion - References

You can vary from this a little if you wish. Indeed we left out "Related Work", which is typically found in Computer Science papers. We suggest discussing any related work in the introduction/methodology, though you are free to add a dedicated "Related Work" section if you wish.

In your methodology, please be clear about how you approached the dataset(s) and pre-processed them before the actual analysis. Also be clear in your report about the type of analysis that you are conducting (remember the type of questions discussed in Lecture 2) Combining analyses is desirable.

You can find the report template in [LaTeX \(https://canvas.uva.nl/courses/59831/files/15868332?wrap=1\)](https://canvas.uva.nl/courses/59831/files/15868332?wrap=1) and [Word \(https://canvas.uva.nl/courses/59831/files/15868349?wrap=1\)](https://canvas.uva.nl/courses/59831/files/15868349?wrap=1). If you are having troubles compiling the LaTeX files in your computer, feel free to [use overleaf \(https://www.overleaf.com/latex/templates/springer-lecture-notes-in-computer-science/kzwwpvhwnvfj#.Wa5y93UjGXy\)](https://www.overleaf.com/latex/templates/springer-lecture-notes-in-computer-science/kzwwpvhwnvfj#.Wa5y93UjGXy). Do not deviate from the template (i.e., do not modify font size, spacing, number of columns).

'Missing' team members and grade distribution among team members

The author list is mandatory. If one (or more) team members did not actually take part at all in the assignment they are to be left off the author list. Emails sent after the submission deadline telling the graders that someone was included on the author list but really wasn't there will be ignored. Authorship as listed on the paper at the submission deadline is what counts. All team members will be assumed to have contributed equally unless a student does not appear on the author list.

Deadline

The **hard deadline** for submission is Monday, September 28, 2026. The Assignment is planned for 2 weeks, but we decided to set the deadline to Monday (instead of Saturday, Sep 26) to accommodate possible last-minute submission issues. We do recommend that you have the Assignment ready to submit Friday and enjoy your weekend. The September 28 deadline is **strict**. No submission means no points, i.e. 0/100. Please note that you can do multiple submissions: a preliminary submission can be made for "safety", and it can be updated until the final deadline.

Software / Videos

In terms of software, you are not restricted to using Python/Pandas/Matplotlib/Scikit-learn as shown in the lectures. However, please understand that if you do use separate software, the TAs can not help you with implementation problems. In previous years students have asked if they can make animations/videos of how data evolved over time. Yes if you make these they can be included as a link in the report. They will be seen, but we would ask that you prioritise the appropriate text and image representations actually in the report.

For any extra question please contact the TA of your Group.

We hope you have fun exploring the UN General Debate corpus and learning more about the SDGs.

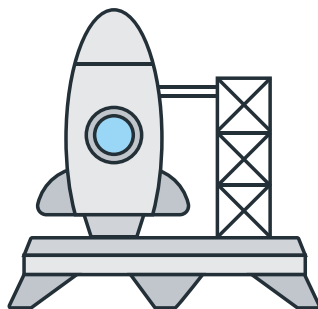
We are very curious to learn about the instigating patterns you'll find in the data and to get to know your creative approaches!

✓ View Rubric

Assignment 1 Rubric	
Criteria	Comments
Question and hypothesis view longer description	Comment
Data preparation view longer description	Comment
Data visualisation view longer description	Comment

Criteria	Comments
Context view longer description	Comment
Model selection and validation view longer description	Comment
Limitations view longer description	Comment
Report quality view longer description	Comment
Creativity view longer description	Comment

Keep in mind, this submission will count for everyone in your Assignment Group group.



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